English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 21.01.2023 Test 2

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Recall that the recurrence $a_n = Aa_{n-1} + B$ *has solution* $a_n = C_1 A^n + C_2$*, where the constants* C_1, C_2 *are determined by the initial conditions.*

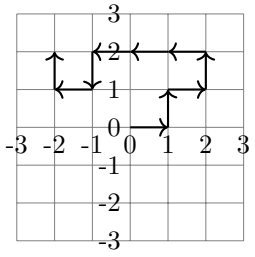
Problem 1. Let X, Y have joint density $f_{X,Y}(x,y) = cx^2 + 1$ for $x, y \geq 0$, $x + 2y \leq 1$, and zero elsewhere, where c is a constant. Find:

1. (0.5 points) c, the density of X and the expectation of Y;
2. (0.25 points) $\mathbb{E}(Y|X = 1/2)$;
3. (0.25 points) the density of $Z = X + 2Y$.

Problem 2. A shop is open from 10:00 to 18:00. Model the times between successive arrivals of customers under 30 as independent exponential variables with mean 5 minutes, and the times between successive customers over 30 as independent exponential variables with mean 10 minutes.

1. (0.5 points) What is the probability that at least 100 customers under 30 visit in one day?
2. (0.25 points) Find the distribution of the waiting time for the first customer of the day.
3. (0.25 points) You may assume that the times between successive customers are independent and have the distribution in part 2[1]. What is the probability that at least 150 people visit in one day?

Problem 3. A point starts at the origin. At each step it chooses one of the coordinate directions uniformly and moves one unit. The figure below shows a sample path of 10 steps.



1. (0.5 points) What is the probability that its x-coordinate exceeds 10 after 1000 steps?
2. (0.5 points) Find the expectation[2] of the squared distance from the origin after 100 steps.

Problem 4. Let $X_0 \sim Exp(1)$ and $X_n = 2X_{n-1} + \epsilon_n$ for $n \in \mathbb{N}$, where the ϵ_n are independent $N(0,1)$ random variables.

1. (0.25 points) Find $\mathbb{E}X_n$ and DX_n.
2. (0.25 points) Let $S_{n,1} = \sum_{i=1}^{n} (X_i - 2X_{i-1})^2$ and $S_{n,2} = \sum_{i=1}^{n} |X_i - 2X_{i-1}|$. Find $\mathbb{E}S_{n,1}$ and $\mathbb{E}S_{n,2}$.
3. (0.5 points) Can you answer parts 1 and 2 when $\epsilon_i \sim N(1,2)$?

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[1] Bonus**: Prove this.

[2] Bonus***: And the variance!